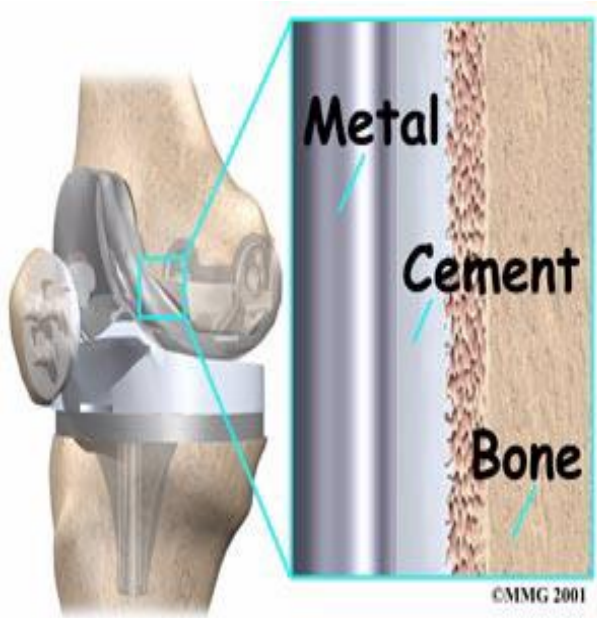
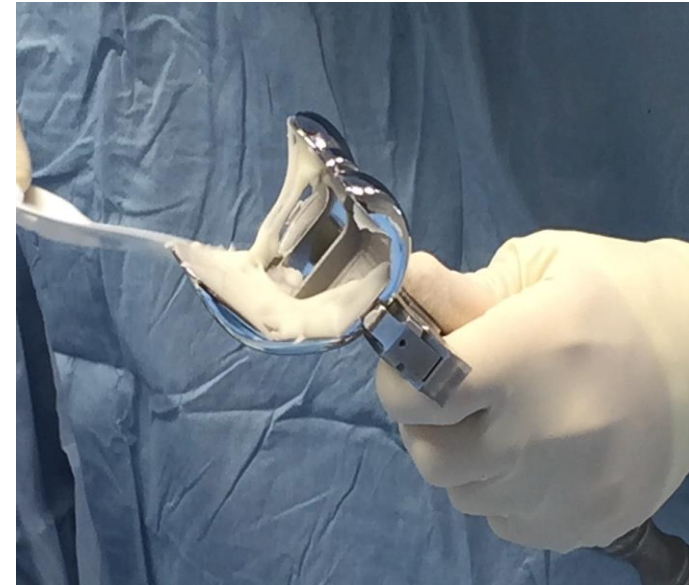


# WHY USE G1A ?



G1A characteristics, such as mechanical performances and working time, allow to this bone cement to be perfectly suitable for Total Knee Arthroplasty Cementation



*“The orthopedic bone cement G1A adopted in arthroplasty implants has shown optimal capacity of fixation of the prosthetic implant with no evident area of bone periprosthetic suffering.” [5]*

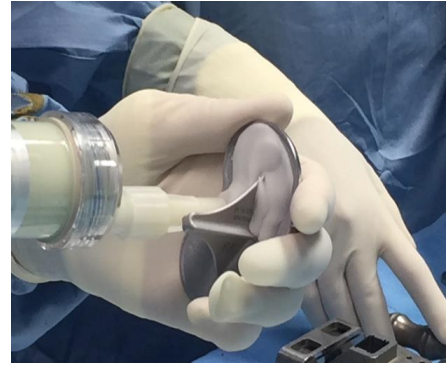
# Orthopaedic Bone Cements- G1A Knee Prosthesis fixation



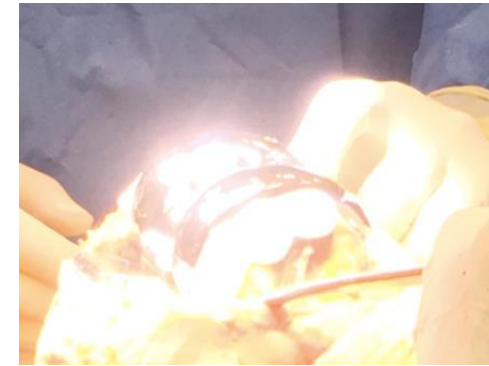
Cementation of tibia with bowl and spatula



Cementation of femoral component with spatula



Cementation of tibia with Picomix Syringe



Implant fixation

## Report clinical and radiologic study [5] on patients undergone total knee prosthesis cemented with G1A<sup>[1]</sup> :

- Clinical and radiologic evaluation with follow up on 2, 6 and 20 months on patients undergone surgery for implant knee arthroplasty by using the bone cement G1A
- PATIENTS: 14 male and 25 female with average age of 70 years old. 21 were right knee and 18 left knee
- RESULTS: No patient subject of the study has highlighted within radiologic exam after 2, 8 and 20 months any evidence mobility of the prosthetic implant. In none of those patients area of peri-prosthetic bone reabsorption has been evident, evidences of articular infection have been highlighted.

# Cementation Technique Tibial component

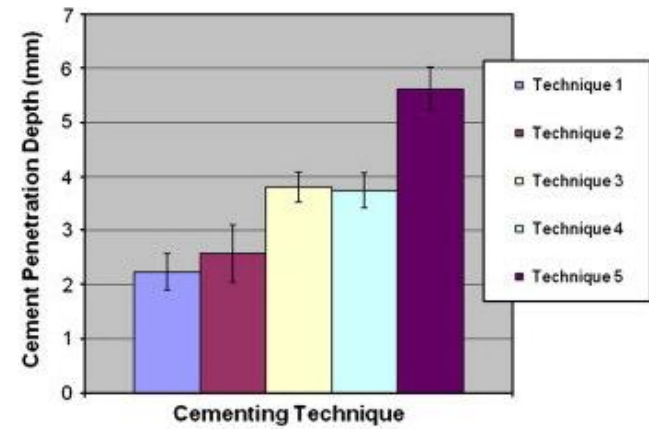
The procedure of prosthesis - fixation, especially the knee, is subjective and varies from surgeon to surgeon.

A study available in literature compare the different performance of 5 different cementation techniques for the cementation of the tibial component [2].

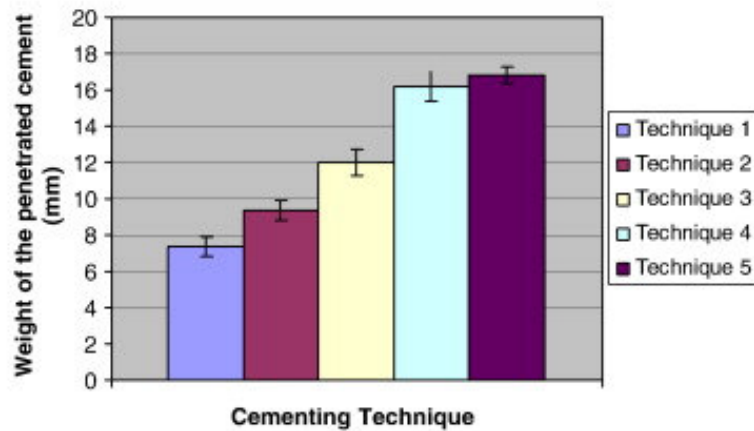
|                                                                                   |                                                                                                                                          |                                                                                      |                                                                                                                                                                |                                                                                     |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|   | <p><b>Technique #1:</b> 10 g of cement was applied in a thin layer on the lower surface of the tibial component.</p>                     |   | <p><b>Technique #2:</b> 20 g of cement was applied in a thin layer on the lower surface of the tibial component.</p>                                           |  |
|  | <p><b>Technique #3:</b> 20 g of cement was applied in equal parts, on both the tibial component and the tibial bone using a spatula.</p> |  | <p><b>Technique #4:</b> 20 g of cement was applied in equal parts on both the tibial component and the tibial bone, but it was finger packed into the bone</p> |                                                                                     |

# Cementation Technique Tibial component

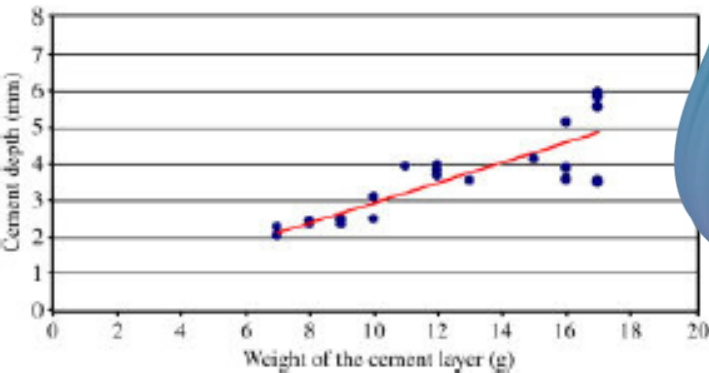
Following, the graphical results of the study:



Graphical results in terms of cement penetration depth



Graphical results in terms of cement penetration depth

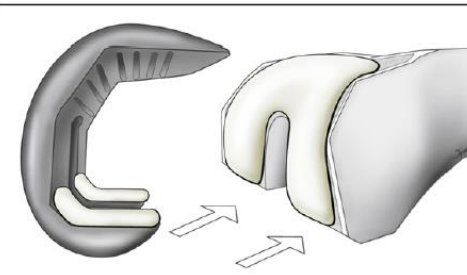
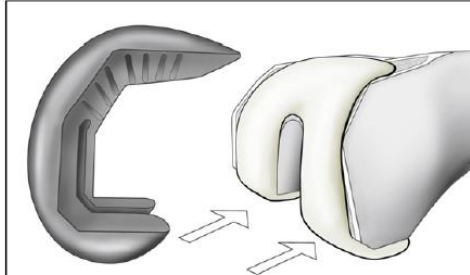
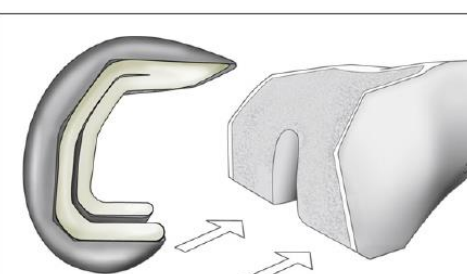
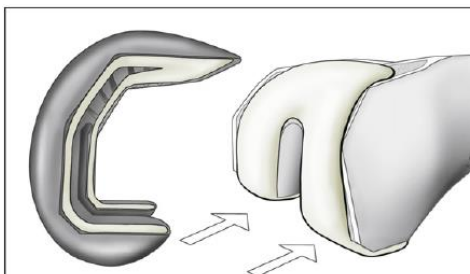


Correlation between calculated weight of cement and the mean penetration depth.

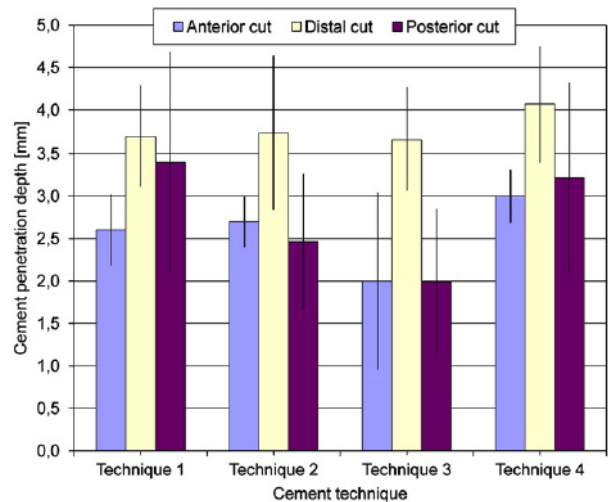
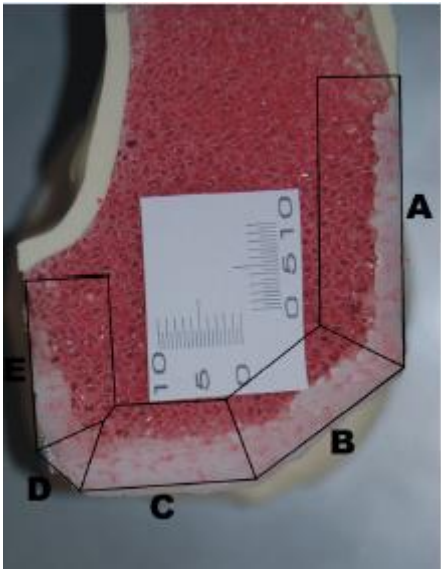
Results: Applying cement to both the undersurface of the tibial baseplate and as well as onto the tibial bone, either by a spatula or finger-packing technique, leads to an optimal cement penetration of 3 to 5 mm[6]. When cement is applied only onto the tibial component, penetration is insufficient.

# Cementation Technique Femoral component

A study available in literature compare the different performance of 4 different cementation techniques for the cementation of the Femoral component [3].

|                                                                                                       |                                                                                                                                                                                                                             |                                                                                                         |                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Technique 1</p>  | <p><b>Technique #1:</b> 37 g of cement was divided in 25 g to cover the anterior and distal femoral cuts with a horseshoe shaped layer and the rest was equally spread over the two posterior flanges of the component.</p> |  <p>Technique 2</p>  | <p><b>Technique #2:</b> 37 g of cement was applied evenly over the entire bone surface. No cement was applied to the implant.</p>                                                          |
|  <p>Technique 3</p> | <p><b>Technique #3:</b> 37 g of cement was applied uniformly on the FC only. No cement was applied on the bone.</p>                                                                                                         |  <p>Technique 4</p> | <p><b>Technique #4:</b> 37 g of bone cement was divided in two equal parts. One half was used to cover the entire femoral bone surface and the other half was spread on the component.</p> |

# Cementation Technique Femoral component with G1A



| Technique | Anterior cut   |               | Distal cut    | Posterior cut  |               |
|-----------|----------------|---------------|---------------|----------------|---------------|
|           | Length (mm)    | Depth (mm)    | Depth (mm)    | Length (mm)    | Depth (mm)    |
| 1         | 31.5<br>(2.17) | 2.6<br>(0.41) | 3.7<br>(0.59) | 16.9<br>(3.38) | 3.4<br>(1.28) |
| 2         | 32.4<br>(1.56) | 2.7<br>(0.30) | 3.7<br>(0.90) | 16.9<br>(2.83) | 2.5<br>(0.79) |
| 3         | 9.8<br>(5.22)  | 2.0<br>(1.04) | 3.7<br>(0.60) | 9.1<br>(4.29)  | 2.0<br>(0.85) |
| 4         | 23.5<br>(3.60) | 3.0<br>(0.31) | 4.1<br>(0.67) | 18.4<br>(2.05) | 3.2<br>(1.11) |

Graphical results of the penetration pattern of the cement in the different sections of the femoral bone, compared by individual cementation technique

Results: Technique 1, where cement was applied onto the anterior and distal bone surfaces, as well as the posterior flanges of the prosthesis, was statistically superior to the other techniques.

Section through the lateral condyle, with superposition of the inner surface of the femoral component. A: Anterior cut; B: anterior chamfer; C: distal cut; D: posterior chamfer; E: posterior cut.

## Bibliography:

[1] Report clinical and radiologic study on patients undergone total knee prosthesis cemented with G1A. 2017- Dr. Castellarin

[2] Cementing the Tibial Component in Total Knee Arthroplasty : Which Technique is the Best? Jan Vanlommel, MD1, , Jean Philippe Luyckx, Ir†, Luc Labey, PhD†, Bernardo Innocenti, PhD†, Ronny De Corte†, Johan Bellemans, MD, PhD1

[3] Cementing the femoral component in total knee arthroplasty: Which technique is the best? Michaël Vaninbroukx a,\*, Luc Labey b, Bernardo Innocenti b, Johan Bellemans a